

ELEC5305 Project Proposal

1. Project Title

Perceptually Guided Speech Enhancement and Spatial Re-Mixing for Complex Stereo Audio Scenes

2. Student Information

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3. Project Overview

The sounds recorded in real environments are often affected by background noise, interference from speakers, and reverberation. Traditional speech enhancement methods can improve signal quality, but they cannot fully utilize the spatial information in stereo recordings. This project aims to develop a MATLAB-based speech enhancement and spatial re-mixing system. This system will analyze the left and right channels in the time-frequency domain, enhance the target speech using spatial cues and speech activity information, and use soft time-frequency masking to suppress interference. Then, HRIR-based binaural processing will be used to re-mix the enhanced speech and background spatially. The main goal is to improve the target speech clarity while maintaining a meaningful spatial listening experience.

4. Background and Motivation

Stereo and binaural recordings contain rich spatial information, which can be used to help distinguish sound sources at different positions. Existing studies have shown that spatial features such as interaural level difference (ILD), interaural phase difference (IPD), and Mixing Vector can reflect the differences between the left and right channels and can be further used to estimate time-frequency masks, thereby improving the effect of stereo speech separation [1], [2].

In addition to noise suppression, speech enhancement systems also need to consider the spatial perception after processing. Van den Bogaert et al. studied a binaural speech enhancement method based on multichannel Wiener filter, and the results

showed that this method could reduce background interference while preserving important binaural spatial cues of the target speech [3]. Li et al. also proposed a binaural speech enhancement method combining interference estimation and Wiener filter, and proved that speech enhancement and binaural cue preservation can be considered simultaneously [4].

Furthermore, HRIR can re-render the sound to different spatial directions through convolution, thereby generating binaural audio with a sense of direction [5]. Therefore, this project hopes to integrate stereo spatial cue analysis, speech enhancement, and HRIR-based spatial re-mixing into the same MATLAB system. Compared with only performing traditional noise suppression, this project not only focuses on improving the clarity of the target speech but also pays attention to the spatial auditory effect of the processed audio.

5. Proposed Methodology

This project will be implemented using MATLAB R2025a. First, controlled stereo audio mixtures will be constructed using clean speech and ambient noise or background speech, and retain the original clean speech as the reference signal for subsequent performance evaluation. If necessary, different left-right channel characteristics or HRIRs will be used to simulate target speech and interference arriving from different spatial directions.

Subsequently, the left and right channels will be converted to the time-frequency domain through Short-Time Fourier Transform (STFT), and stereo spatial cues such as ILD and IPD will be extracted. These features can reflect the spatial differences of different sound sources in the left and right channels and can be used to assist time-frequency masking [1], [2].

In the speech enhancement stage, the system will use a simple Voice Activity Detection (VAD) method to distinguish speech-active and noise-dominant regions, and estimate the spectral characteristics of the background interference. A Wiener-style soft mask will be implemented first as a baseline, and then stereo spatial weighting will be added on top of it. The time-frequency regions that are more consistent with the spatial features of the target speech will receive higher weights, while the regions that are more likely to belong to background interference will be further suppressed, thereby improving the clarity of the target speech [3], [4].

Finally, the enhanced target speech will be reconstructed through inverse STFT, and the approximate background components will be obtained using complementary masks. Then, HRIR convolution will be performed on the target speech and background sounds for spatial re-mixing, allowing them to be repositioned in different spatial directions, generating a binaural output with spatial cues [5].

The system performance will be evaluated mainly through SNR improvement and spectrogram comparison; if time permits, SDR or simple perceptual evaluation will also be considered.

6. Expected Outcomes

This project is expected to complete a functional MATLAB prototype for enhancing the target speech in complex stereo audio and generating spatially mixed stereo or binaural output. The system should be able to improve the clarity of the target speech while controlling the spatial positions of the speech and background sounds.

The project will generate the enhanced audio, time-frequency masks, spectrograms, and related evaluation results. The system performance will be mainly evaluated by SNR improvement, and SDR or PESQ may be included if conditions permit. The final outcome will also include MATLAB source code, example audio, evaluation charts, final report, and complete GitHub documentation.

7. Timeline

Weeks	Planned Work
1–2	Define project scope and complete initial research
3–5	Literature review, dataset selection and preliminary STFT experiments
6–7	Implement VAD and baseline Wiener speech enhancement
8–9	Implement stereo cue analysis and stereo-guided masking
10	Integrate stereo processing and time-frequency masking
11	Implement HRIR-based spatial re-mixing
12	Evaluation, optimisation and optional extensions
13	Final report, demonstration and GitHub documentation

8. References

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